

Phase Descent: Identifying Error Lineage in Passive Correlators from Harmonic Phase Coherence

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Abstract

Passive correlators recover signal overlap from the energy imbalance between complementary interference channels — a physical realization of the polarization identity. When the projections are phase-like, the output is cosine in the angular difference between detector settings, and any distortion generated by a single latent perturbation through a single nonlinearity produces angular Fourier harmonics satisfying $\phi_n - n\phi_1 = 0 \pmod{\pi}$. We call this property *phase descent*; it follows from the Chebyshev expansion of analytic functions on $[-1, 1]$ but is, to our knowledge, new as a fault-lineage diagnostic. Phase descent partitions systematic errors into two structurally distinct classes — analyzer asymmetry produces phase-locked odd harmonics, while recombiner asymmetry produces phase-incoherent even harmonics plus a DC offset — enabling a calibration diagnostic that extracts fault type, fault severity, and its own validity bound from a single angular sweep. Numerical experiments confirm phase-locking to within 10^{-4} rad for analyzer errors up to 20% severity, while recombiner-induced inter-harmonic phases drift continuously with error magnitude. The underlying principle applies naturally to any measurement architecture in which the observable is well modeled as $f(\cos(\theta - \theta_0))$.

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1 Introduction

Passive correlators that measure the overlap between two signals by comparing the energy in complementary interference channels appear across quantum optics [1], [2], [3], polarimetry [4], [5], and CMB observations [6], [7]. These instruments share a common mathematical skeleton: two local projections of a shared field are combined into bright (sum) and dark (difference) channels, and the normalized contrast

$$E(a, b) = \frac{P_{\text{Br}} - P_{\text{Dk}}}{P_{\text{Br}} + P_{\text{Dk}}}$$

serves as the correlation observable. When the instrument is symmetric, the output is $E(a, b) = \cos(a - b)$, where a and b are the local analyzer settings.

The normalization step introduces a rational-function nonlinearity whose error-propagation properties have not been fully characterized spectrally. While general harmonic-distortion frameworks exist for nonlinear systems [8], [9], they do not exploit the specific structure of the cosine observable that emerges from normalized bright/dark contrast. This paper develops a framework that does. The central result is a *phase-descent invariant*: any distortion arising from a single parametric perturbation passing through a single nonlinearity produces angular Fourier harmonics whose phases satisfy $\phi_n - n\phi_1 = 0 \pmod{\pi}$, regardless of the severity or functional form of the nonlinearity. As shown in Section 2, this is a direct consequence of the Chebyshev expansion of analytic functions on $[-1, 1]$ [10], [11]; the contribution of the present paper is to develop it as a *diagnostic* for fault lineage in normalized-contrast instruments.

This invariant partitions systematic errors into two structurally distinct classes — *analyzer asymmetry* (phase-locked odd-harmonic descendants) and *recombiner asymmetry* (phase-incoherent even harmonics plus DC offset) — enabling a calibration diagnostic that extracts fault type, fault severity, and its own validity bound from a single angular sweep.

The framework arose from the analysis of a passive transformer-coupled correlator designed to reproduce cosine overlap from normalized bright/dark power contrast. That architecture produces $E(a, b) = \cos(a - b)$ with high fidelity under symmetry, but the analysis of how different symmetry violations manifest spectrally revealed the phase-descent structure and the two-family error taxonomy presented here.

The paper is organized as follows. Section 2 establishes the mathematical framework: the phase-descent invariant, its converse, and the polarization identity that underlies the measurement principle. Section 3 instantiates the framework in a concrete passive correlator model. Section 4 presents numerical validation. Section 5 formulates the complete diagnostic pipeline. Section 6 discusses implications and limitations. Section 7 concludes.

2 Mathematical Framework

2.1 The Polarization Identity as Measurement Principle

The bright/dark correlator rests on the polarization identity for complex inner-product spaces [12], [13]. For complex signals x_A and x_B ,

$$|x_A + x_B|^2 - |x_A - x_B|^2 = 4 \operatorname{Re}(x_A \overline{x_B}). \quad (1)$$

Normalizing by total energy,

$$|x_A + x_B|^2 + |x_A - x_B|^2 = 2(|x_A|^2 + |x_B|^2),$$

yields the normalized contrast

$$E = \frac{|x_A + x_B|^2 - |x_A - x_B|^2}{|x_A + x_B|^2 + |x_A - x_B|^2} = \frac{2 \operatorname{Re}(x_A \overline{x_B})}{|x_A|^2 + |x_B|^2}.$$

When the projections are phase-like — $x_A = e^{-ia}$, $x_B = e^{-ib}$ — this reduces to

$$E = \operatorname{Re}(e^{-ia} e^{+ib}) = \cos(a - b).$$

Thus, any instrument that performs coherent sum/difference recombination followed by quadratic (power) detection and normalization is computing an inner product. The cosine structure is not a model assumption; it is a consequence of the measurement architecture.

2.2 The Phase-Descent Invariant

The key spectral fact we exploit is a direct consequence of the Chebyshev expansion of analytic functions on $[-1, 1]$ [10], [11]. We state it as a lemma, then extract the diagnostic-relevant invariant as a corollary.

Lemma (Chebyshev consequence). Let $g(\theta) = f(\cos(\theta - \theta_0))$ where f is analytic on $[-1, 1]$ (i.e., admits a power series convergent on a neighborhood of $[-1, 1]$). Then g has a Fourier cosine expansion

$$g(\theta) = \sum_{n \geq 0} c_n \cos(n(\theta - \theta_0)),$$

with all $c_n \in \mathbb{R}$.

Proof sketch. The Chebyshev polynomials of the first kind satisfy $T_n(\cos \psi) = \cos(n\psi)$; equivalently, each power $\cos^k \psi$ decomposes as a finite real linear combination of $\{\cos(m\psi)\}_{0 \leq m \leq k}$ of the same parity as k . Since f is analytic on $[-1, 1]$, its power series $\sum a_k u^k$ converges uniformly on $[-1, 1]$, and the resulting cosine series for g may be reorganized term-by-term. The full coefficient calculation is given in Appendix A. $\square$

Corollary (Phase descent). Under the same hypotheses, the complex Fourier coefficients H_n of g have phases satisfying

$$\phi_n - n\phi_1 = 0 \pmod{\pi}. \quad (2)$$

Proof. The n -th harmonic is $c_n \cos(n(\theta - \theta_0))$, so $\phi_n = n\theta_0$ when $c_n > 0$ and $\phi_n = n\theta_0 + \pi$ when $c_n < 0$. In either case, $\phi_n - n\phi_1 \equiv 0 \pmod{\pi}$. $\square$

We refer to (Equation 2) as the *phase-descent invariant*. It is *structural*: it holds regardless of the form of f , the expansion coefficients, or the severity of the perturbation. While the underlying Chebyshev expansion is classical, our use of (Equation 2) as a diagnostic for fault lineage in normalized-contrast instruments — developed in the remainder of this paper — is, to our knowledge, new.

2.3 The Converse: Phase-Lock Violation as Multi-Source Indicator

Proposition. Suppose

$$\delta(\theta) = f_1(\cos(\theta - \theta_1)) + f_2(\cos(\theta - \theta_2))$$

with f_1, f_2 analytic on $[-1, 1]$ and $\theta_1 \not\equiv \theta_2 \pmod{\pi}$. Then there exist harmonic orders n for which $\phi_n - n\phi_1 \not\equiv 0 \pmod{\pi}$, except on a measure-zero subset of the coefficient space.

Proof. By the Lemma applied to each summand, the n -th complex Fourier coefficient is

$$H_n = c_n^{(1)} e^{in\theta_1} + c_n^{(2)} e^{in\theta_2}, \quad c_n^{(1)}, c_n^{(2)} \in \mathbb{R} \cup \{\pm i\mathbb{R}\}_{(\text{sign-ambiguous})}.$$

For H_n to satisfy $\phi_n \equiv n\theta_0 \pmod{\pi}$ for *some* common θ_0 across all n , the two complex contributions must be parallel in the complex plane at every harmonic order — equivalently, the ratio $c_n^{(1)} e^{in\theta_1} / c_n^{(2)} e^{in\theta_2}$ must be real for every n with $H_n \neq 0$. This is a countable family of algebraic constraints on the coefficients $\{c_n^{(1)}, c_n^{(2)}\}$, each of codimension one in the parameter space of analytic perturbations; their joint satisfaction is therefore measure-zero unless $\theta_1 \equiv \theta_2 \pmod{\pi}$, in which case both contributions share a common phase origin and the proposition’s hypothesis is violated. $\square$

A concrete witness: take $f_1(u) = \epsilon_1 u$ and $f_2(u) = \epsilon_2 u^2$ with $\theta_1 = 0$, $\theta_2 = \pi/4$. Then $H_1 = (\epsilon_1/2)e^{i \cdot 0}$ has phase $\phi_1 = 0$, while H_2 contains a $(\epsilon_2/4)e^{i\pi/2}$ term — giving $\phi_2 = \pi/2$ — so $\phi_2 - 2\phi_1 = \pi/2 \not\equiv 0 \pmod{\pi}$. Phase descent fails after one harmonic.

Together, the corollary and its converse provide a *necessary* condition for single-source origin: phase descent must hold if the distortion is generated by a single latent phase variable through a single nonlinearity. Its violation rules out that single-origin description and constitutes evidence for multi-origin or structurally richer distortion.

Remark. The “measure-zero” caveat above identifies a genuine spectral indistinguishability: if two independent sources happen to share a phase origin ($\theta_1 \equiv \theta_2 \pmod{\pi}$), their superposition lies in the kernel of the test. In practice, the phase-descent test diagnoses the number of *distinct phase origins*, not the number of physical sources per se. Two sources that enter at the same angular location are spectrally indistinguishable from one.

3 Physical Instantiation: The Bright/Dark Channel Correlator

3.1 Architecture

We instantiate the framework in a concrete passive correlator: a transformer-coupled network driven by a balanced three-phase source at angular frequency ω . The network consists of nine magnetically coupled coils organized in a staged cascade:

Table 1: Coil indexing and function in the nine-coil correlator network.

Stage	Coils	Function
Source	p_a, p_b, p_c (indices 0–2)	Three-phase voltage drive
Clarke pickup	α, β (indices 3–4)	Clarke-transform decomposition
Analyzer	A, B (indices 5–6)	Angle-dependent projections
Recombiner	Br, Dk (indices 7–8)	Bright/dark interference channels

The magnetic part of the network satisfies the coupled-inductor equation

$$\mathbf{v} = \mathbf{R}_s \mathbf{i} + \mathbf{L} \dot{\mathbf{i}},$$

where $\mathbf{L}$ is a 9×9 symmetric positive-definite mutual-inductance matrix and $\mathbf{R}_s$ is the diagonal series-resistance matrix.

The source-to-Clarke coupling implements the Clarke transform weights [14]:

$$\mathbf{M}_{pC} = k_C \begin{bmatrix} 1 & -1/2 & -1/2 \\ 0 & \sqrt{3}/2 & -\sqrt{3}/2 \end{bmatrix}.$$

The Clarke-to-analyzer stage uses angle-dependent couplings:

$$A \sim \cos a \alpha + \sin a \beta, \quad B \sim \cos b \alpha + \sin b \beta.$$

The bright/dark recombiner uses signed couplings:

$$\text{Br} \sim A + B, \quad \text{Dk} \sim A - B,$$

where the minus sign is carried by the dark-branch mutual inductance.

3.2 Steady-State Phasor Solution

At steady state, the phasor equation is

$$(\mathbf{R}_s + j\omega\mathbf{L} + \mathbf{Z}_{\text{term}})\mathbf{I} = \mathbf{V}_{3\phi},$$

where $\mathbf{Z}_{\text{term}}$ collects the intermediate-port resistive loads (R_{kL} on each Clarke and analyzer port) and the tank shunt impedances ($Z_{\text{Br}} = 1/Y_{\text{Br}}$, $Z_{\text{Dk}} = 1/Y_{\text{Dk}}$), with

$$Y_{\text{Br}} = \frac{1}{R_{\text{Br,sh}}} + j\omega C_{\text{Br}}, \quad C_{\text{Br}} = \frac{1}{\omega^2 L_{\text{Br}}},$$

so that each tank is tuned to resonance at the source frequency.

The correlator output is

$$E(a, b) = \frac{P_{\text{Br}} - P_{\text{Dk}}}{P_{\text{Br}} + P_{\text{Dk}}}, \quad P_{\text{Br}} = \frac{|V_{\text{Br}}|^2}{R_{\text{Br,sh}}}, \quad P_{\text{Dk}} = \frac{|V_{\text{Dk}}|^2}{R_{\text{Dk,sh}}}.$$

3.3 Error Taxonomy

Two classes of asymmetric perturbation are introduced.

Analyzer asymmetry (ϵ_A): unequal α/β coupling gains,

$$k_{A,\alpha} = k_A(1 + \epsilon_A), \quad k_{A,\beta} = k_A(1 - \epsilon_A).$$

This enters as a single perturbation modifying the effective projection angle, passing through the rational normalization nonlinearity. By the phase-descent corollary, the resulting error signal inherits a phase-coherent harmonic structure.

Recombiner asymmetry (ϵ_R): branch-local coupling imbalance,

$$k_{Br,A} = k_R(1 + \epsilon_R), \quad k_{Br,B} = k_R(1 - \epsilon_R),$$

applied to one branch (bright-only or dark-only). Because bright and dark channels contribute to the normalized ratio through separate terms, the resulting error signal is not well described as a single phase variable through a single nonlinearity. By the converse proposition, the resulting harmonics need not satisfy the phase-descent relation.

The physical distinction is this: analyzer asymmetry acts *before* the bright/dark recombination. It deforms the local projection of the shared field — effectively rotating the analyzer coordinate — so that all downstream error flows from a single distorted angular variable into the normalization nonlinearity. That is why the error harmonics are descendants of one parent. Recombiner asymmetry, by contrast, acts *after* the analyzer channels already exist. It alters the coupling weights differently in the bright and dark branches, so the numerator ($P_{Br} - P_{Dk}$) and denominator ($P_{Br} + P_{Dk}$) absorb perturbations with different effective phase origins. The normalized output is therefore a mixture of contributions that do not share a single angular lineage, and the inter-harmonic phase coherence that characterizes the analyzer pathway is absent.

Table 2: Spectral signatures of the two fault families.

Property	Analyzer asymmetry	Recombiner asymmetry
Harmonics	Odd ($n = 1, 3, 5, \dots$)	DC + even ($n = 0, 2, 4, \dots$)
Phase relation	$\phi_3 - 3\phi_1 \approx 0$ or π	No invariant
Diagnostic test	Phase-lock: $\text{dist}(\Delta\phi_{13}, \{0, \pi\}) < \tau$	Phase drift: $\phi_2 - 2\phi_1$ varies with ϵ_R
Origin	Pre-recombination: one distorted angle $\rightarrow$ one nonlinearity	Post-analyzer: separate branch perturbations $\rightarrow$ multiple effective origins
Severity indicator	$ H_3 $	$ H_2 $
Validity monitor	$ H_5 / H_3 $	—

4 Numerical Validation

4.1 Angular Fidelity of the Symmetric Correlator

We first verify that the symmetric ($\epsilon_A = \epsilon_R = 0$) loaded-port correlator reproduces $E(a, b) = \cos(a - b)$ with high fidelity.

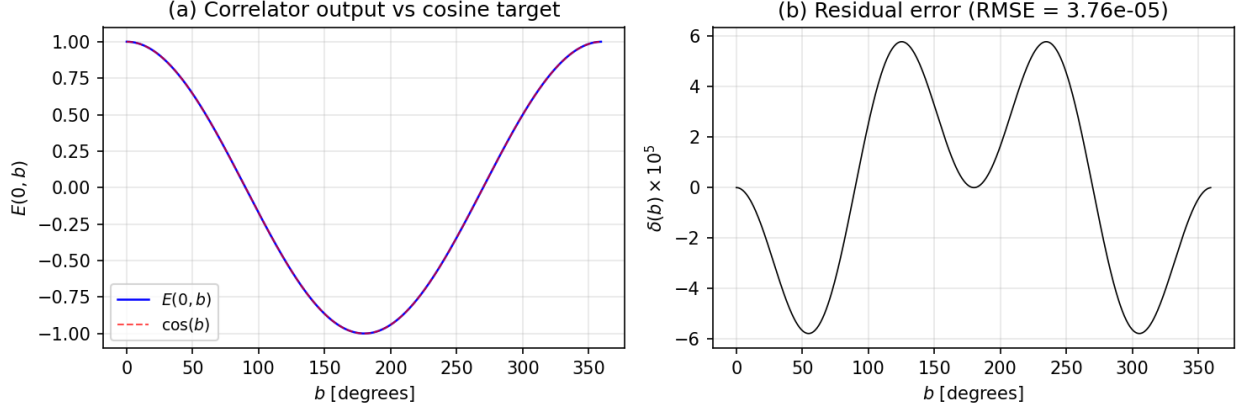


Figure 1: Angular fidelity of the symmetric correlator. (a) $E(0, b)$ compared with $\cos(b)$. (b) Residual error $\delta(b) = E(0, b) - \cos(b)$, showing RMSE at the 10^{-5} level.

RMSE = 3.755172e-05

Max |error| = 5.781366e-05

Min eigenvalue of L = 3.779951e-04

The loaded-port model ($R_{kL} = 1000 \Omega$) reproduces the cosine target with $\text{RMSE} \approx 3.8 \times 10^{-5}$ and maximum absolute error $\approx 5.8 \times 10^{-5}$. The residual error surface is featureless, consistent with the rotational symmetry of the nominal coupling construction.

4.2 Analyzer Asymmetry: Phase-Locked Odd Harmonics

We now introduce analyzer asymmetry ϵ_A and examine the angular error spectrum.

The results confirm the phase-descent prediction:

1. Analyzer asymmetry produces a dominant odd-harmonic family ($n = 1, 3, 5$) with negligible even content.
2. The phase-lock quantity $\text{dist}(\phi_3 - 3\phi_1, \{0, \pi\})$ remains at $\sim 10^{-4}$ rad across the full range from 1% to 20% asymmetry.
3. The leakage ratio $|H_5|/|H_3|$ grows monotonically with severity, from $\sim 5 \times 10^{-3}$ at 1% to $\sim 8 \times 10^{-2}$ at 20%, providing an in-band validity monitor.

4.3 Recombiner Asymmetry: Phase-Incoherent Even Harmonics

We repeat the spectral analysis for branch-local recombiner asymmetry, applied to the bright branch and dark branch independently.

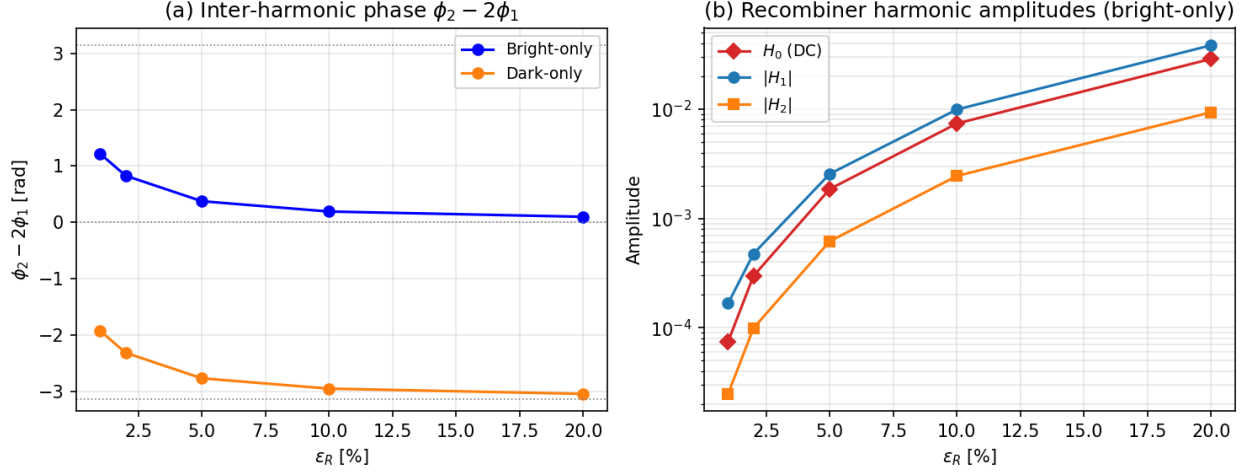


Figure 2: Recombiner asymmetry phase analysis. (a) Inter-harmonic phase $\phi_2 - 2\phi_1$ for bright-only (blue) and dark-only (orange) recombiner perturbations, showing continuous drift with ϵ_R — no invariant. (b) Harmonic amplitudes for the bright-only case, showing DC (H_0), H_1 , and H_2 content.

Recombiner inter-harmonic phase ($\phi_2 - 2\phi_1$):

eps_R	Bright	Dark
1%	1.2191	-1.9225
2%	0.8293	-2.3123
5%	0.3760	-2.7656
10%	0.1932	-2.9484
20%	0.0998	-3.0418

In sharp contrast with the analyzer case:

1. Recombiner asymmetry produces DC, $n = 1$, and $n = 2$ content, with negligible odd harmonics.
2. The inter-harmonic phase $\phi_2 - 2\phi_1$ drifts continuously with ϵ_R , with bright-only and dark-only branches approaching different asymptotic values — confirming that these harmonics do not descend from a single phase origin.
3. The absence of a phase invariant is the diagnostic signature: these harmonics are *siblings*, not *descendants*.

The asymptotic values $\phi_2 - 2\phi_1 \rightarrow 0$ (bright) and $\rightarrow -\pi$ (dark) are not coincidental. With bright-only asymmetry, the bright-channel voltage at the recombiner becomes $V_{\text{Br}} = V_0 + \epsilon_R V_d$, where V_0 and V_d are the symmetric bright and dark phasors respectively. At resonance these phasors are nearly orthogonal in the complex plane, so the cross term $2\epsilon_R \text{Re}(V_0^* V_d)$ vanishes to leading order and the power perturbation is $O(\epsilon_R^2)$:

$$\delta P_{\text{Br}} \approx \epsilon_R^2 |V_d|^2 = \epsilon_R^2 P_{\text{Dk}}^{(0)}.$$

Substituting $P_{\text{Br}}^{(0)} = 4 \cos^2(b/2)$, $P_{\text{Dk}}^{(0)} = 4 \sin^2(b/2)$ into the linearization $\delta E = (2P_{\text{Dk}}^{(0)}/(P_{\text{Br}}^{(0)} + P_{\text{Dk}}^{(0)})^2) \delta P_{\text{Br}}$ and expanding $\sin^4(b/2)$ in cosines yields

$$\delta E_{\text{bright}}(b) = \frac{3}{4} \epsilon_R^2 - \epsilon_R^2 \cos b + \frac{1}{4} \epsilon_R^2 \cos 2b + O(\epsilon_R^3).$$

Hence $H_1 < 0$ (so $\phi_1 = \pi$) and $H_2 > 0$ (so $\phi_2 = 0$), giving $\phi_2 - 2\phi_1 \equiv 0 \pmod{2\pi}$.

The dark-only case is symmetric: $V_{\text{Dk}} = V_d + \epsilon_R V_0$, and the same expansion with $\cos^4(b/2)$ in place of $\sin^4(b/2)$ gives

$$\delta E_{\text{dark}}(b) = -\frac{3}{4}\epsilon_R^2 - \epsilon_R^2 \cos b - \frac{1}{4}\epsilon_R^2 \cos 2b + O(\epsilon_R^3),$$

so $H_1 < 0$ ($\phi_1 = \pi$) and $H_2 < 0$ ($\phi_2 = \pi$), giving $\phi_2 - 2\phi_1 \equiv -\pi \pmod{2\pi}$.

The asymptotic phase difference between branches is therefore exactly π — a fingerprint of which branch carries the asymmetry. The drift away from these asymptotes at small ϵ_R in Figure 2(a) reflects the relative magnitude of the cross-term $2\epsilon_R \text{Re}(V_0^* V_d)$, which is small but nonzero off resonance and becomes negligible relative to the $\epsilon_R^2 |V_{0,d}|^2$ terms only as ϵ_R grows.

4.4 Mixed-Case Recovery

In practice, both error sources may coexist. We test the diagnostic pipeline on a grid of simultaneous analyzer and recombiner asymmetries.

The extraction procedure is:

1. Estimate $\hat{\epsilon}_A$ from $|H_3|$ (exclusive to the analyzer pathway).
2. Confirm attribution via $\phi_3 - 3\phi_1$.
3. Project H_1 onto the analyzer-consistent phase direction.
4. Estimate $\hat{\epsilon}_R$ from $|H_2|$ (primarily recombiner).
5. Monitor validity via $|H_5|/|H_3|$.

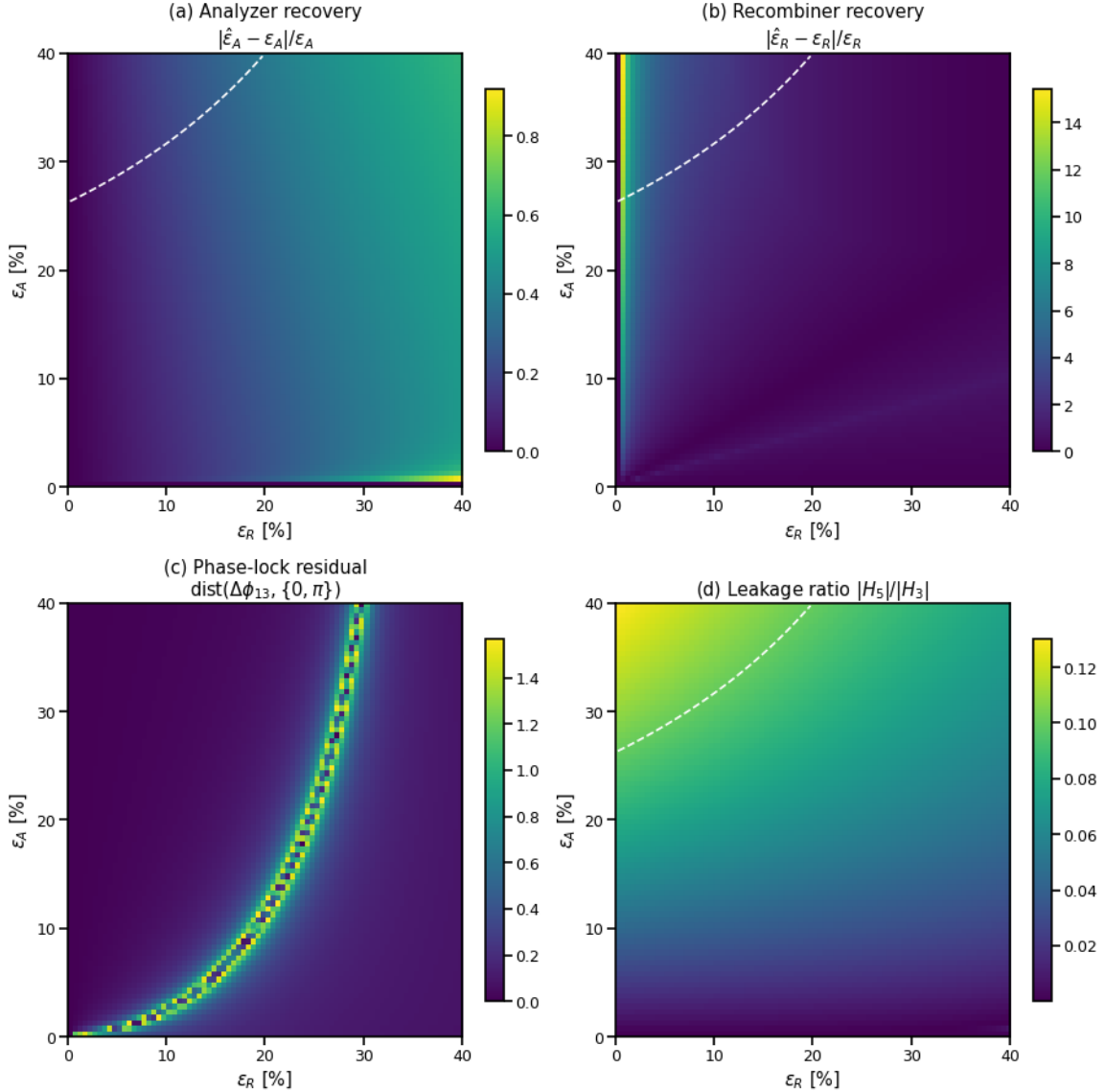


Figure 3: Mixed-case recovery accuracy over the (ϵ_A, ϵ_R) grid. (a) Analyzer recovery relative error. (b) Recombiner recovery relative error. (c) Phase-lock residual. (d) Leakage ratio $|H_5|/|H_3|$.

The mixed-case grid reveals the operational boundaries of the pipeline:

- **Analyzer recovery** is robust: $|H_3|$ -based estimates remain accurate across most of the grid because H_3 is largely exclusive to the analyzer pathway.
- **Recombiner recovery** from $|H_2|$ alone is reliable in the pure and dominant-recombiner regimes, but degrades when analyzer asymmetry dominates, because the normalization denominator carries analyzer-induced b -dependence that bleeds into H_2 .
- **Phase-lock stability** degrades gradually in the mixed regime but remains a useful trust metric.
- **Leakage ratio** grows primarily with ϵ_A and is weakly affected by ϵ_R , confirming it as a valid monitor for the analyzer pathway.

The dashed white contour overlaid on panels (a), (b), and (d) marks the trust-monitor threshold $\Lambda_{53} = 0.1$ derived analytically in Step 5 of §5. The contour traces $\epsilon_A \approx 30\%$ across the ϵ_R axis, and panels (a) and (b) show that recovery error begins to grow appreciably outside this boundary — the leakage ratio thus operationalizes the perturbative-validity bound, providing a directly observable warning before the linear decomposition fails silently. This is the operational payoff of having a built-in trust monitor: a user running the diagnostic on real data sees Step 5 trip *before* trusting attribution from a regime where Steps 1–4 are no longer valid.

The exact pixel values in Figure 3 depend on the network parameters of this particular correlator (Appendix B), but the four qualitative features just listed — analyzer-recovery flatness in the ϵ_R direction, recombiner-recovery degradation along ϵ_A , the interior phase-lock failure ridge, and leakage growth along ϵ_A — are structural consequences of the perturbative derivations in §4.2–4.3 and the diagnostic decomposition of §5, not of the specific physical instantiation. Any normalized-contrast correlator analyzed by the same pipeline should produce qualitatively similar heatmaps, with quantitative magnitudes depending on its own parameters.

5 The Diagnostic Pipeline

The preceding numerical experiments validate a complete calibration diagnostic that operates in five steps from a single angular sweep.

Step 1 — Spectral triage. Compute the Fourier spectrum of the angular error $\delta(b) = E(0, b) - \cos(b)$. The presence of H_3 signals an analyzer candidate; the presence of H_2 plus DC signals a recombiner candidate.

Step 2 — Phase-invariant test. Compute $\Delta\phi_{13} = \text{wrap}(\phi_3 - 3\phi_1)$. If $\text{dist}(\Delta\phi_{13}, \{0, \pi\}) < \tau$ for a chosen threshold τ (empirically, $\tau \sim 10^{-2}$ rad suffices), the analyzer attribution is confirmed by the phase-descent corollary.

Step 3 — Recombiner characterization. Compute $\Delta\phi_{21} = \text{wrap}(\phi_2 - 2\phi_1)$. The sign encodes branch identity (bright vs. dark); the magnitude and drift with ϵ_R encode severity. The absence of a phase invariant is itself the diagnostic: these harmonics are siblings from independent pathways, not descendants of a single nonlinearity.

Step 4 — Mixed-case decomposition. When both error sources are suspected, project H_1 onto the analyzer-consistent phase direction:

$$H_1^{(A)} = |H_1| \cos(\phi_1 - \phi_3/3) e^{j\phi_3/3}, \quad H_1^{(R)} = H_1 - H_1^{(A)}.$$

This decomposes the fundamental into an analyzer-attributed component (phase-locked to H_3) and a residual assigned to the recombiner pathway. The decomposition is first-order accurate in the perturbation parameters; the leakage ratio in Step 5 monitors this approximation.

Step 5 — Validity metric. The leakage ratio $\Lambda_{53} = |H_5|/|H_3|$ monitors whether the low-order approximation remains trustworthy. The threshold $\Lambda_{53} \approx 0.1$ is not arbitrary. Expanding the normalized contrast $E(0, b)$ in ϵ_A around the symmetric operating point and reading off the odd harmonics yields, to leading order, $|H_3| = O(\epsilon_A^2)$ and $|H_5| = O(\epsilon_A^4)$, so

$$\Lambda_{53} = \frac{|H_5|}{|H_3|} = O(\epsilon_A^2)$$

in the perturbative regime. Setting $\Lambda_{53} = 0.1$ corresponds to $\epsilon_A \sim 0.3$ — close to the radius beyond which the rational normalization $1/(P_{\text{Br}} + P_{\text{Dk}})$ ceases to be well-approximated by its first-order Taylor expansion in ϵ_A near the symmetric point. The numerical sweep (§4.2, $\Lambda_{53} \in [5 \times 10^{-3}, 8 \times 10^{-2}]$ for $\epsilon_A \in [1\%, 20\%]$) is consistent with this scaling, and confirms that $\Lambda_{53} \lesssim 0.1$ is a defensible operating bound for the linear decomposition in Step 4. Figure 3(d) traces the $\Lambda_{53} = 0.1$ contour across the mixed-case grid, and panels 3(a, b) show that recovery error grows precisely outside this contour — visual confirmation that the trust monitor fires at the perturbative-validity boundary derived above. When Λ_{53} exceeds ~ 0.1 , the analyzer harmonic ladder has significant higher-order content and the linear decomposition loses accuracy; users should fall back to direct nonlinear estimation rather than trust the Step-4 projection.

All five steps operate on the Fourier coefficients of a single angular sweep — the same measurement already required to characterize the correlation curve. No additional waveforms or hardware are needed.

6 Discussion

6.1 The Phase-Descent Invariant as a Structural Discriminator

The central result of this paper is not the specific error taxonomy of the correlator studied, but the underlying principle: *phase coherence among harmonics diagnoses a single generating degree of freedom, while its violation rules out a single-origin description.*

A correlation curve generated by a single constrained perturbation passing through a single nonlinearity — whether rational normalization, saturation, or any other map with convergent power series — produces harmonics that are children of one parent. The phase-descent relation $\phi_n - n\phi_1 = 0 \pmod{\pi}$ is the algebraic signature of that lineage. Its violation rules out description by a single latent phase variable through a single nonlinearity, and constitutes evidence for multi-origin or structurally richer distortion.

This principle should apply naturally to any measurement architecture in which the observable is well modeled as $f(\cos(\theta - \theta_0))$, including interferometric fringe measurements with systematic path-length errors, polarimetric instruments with differential gain or rotation errors [4], [5], and normalized-contrast instruments in quantum optics [1], [2]. The specific two-family taxonomy derived here is particular to the bright/dark correlator architecture; the phase-descent test itself is not. The same observation extends to the mixed-case recovery surfaces of Figure 3: while the absolute pixel values reflect this network’s parameters, the qualitative pattern — analyzer-vs-recombiner separation in panels (a, b), interior phase-lock failure in (c), ϵ_A -driven leakage in (d) — should be reproducible on any normalized-contrast correlator analyzed by the same pipeline.

Relative to general harmonic-distortion frameworks for nonlinear systems [8], [9], the contribution here is to identify a phase-coherence invariant that depends only on the cosine structure of the observable — not on the form of the nonlinearity — and to use it as a single-vs-multi-origin discriminator. Compared to existing self-consistent calibration methods for polarimeters [5], which assume a parametric Mueller-matrix model and recover its parameters by least squares, the phase-descent test is model-free in the sense that it diagnoses the *number of generating degrees of freedom* before any parametric model is fit.

6.2 Connection to the Polarization Identity

The two numbered equations of this paper occupy different layers of the same structure. The polarization identity (Equation 1),

$$|x_A + x_B|^2 - |x_A - x_B|^2 = 4 \operatorname{Re}(x_A \overline{x_B}),$$

explains *what the instrument computes when it works* — it is why the correlator produces cosine. The phase-descent invariant (Equation 2),

$$f(\cos(\theta - \theta_0)) \Rightarrow \phi_n - n\phi_1 = 0 \pmod{\pi},$$

explains *how to identify what went wrong when it doesn’t* — it is how distortions reveal their lineage. The polarization identity guarantees the cosine output; phase descent extracts diagnostic information from distortions of that cosine. They are parent and child: the identity is the deeper mathematical fact, but the invariant is the operationally new contribution.

6.3 Limitations and Open Questions

The framework assumes:

- **Analytic nonlinearity.** The function f must be analytic on $[-1, 1]$ — that is, it must admit a convergent power series. The proof relies on term-by-term expansion; merely smooth (i.e., C^∞ but non-analytic) functions are not covered.
- **Single-parameter perturbation.** Each fault family is assumed to be generated by one degree of freedom. Multiple simultaneous perturbations of the same family (e.g., two different analyzer misalignments) would require extension.

- **Stationarity.** The analysis assumes a steady-state angular sweep. Time-varying or stochastic errors would require ensemble averaging.

The mixed-case recovery results reveal a concrete limitation: the normalization denominator couples the two error pathways, so that H_2 alone is not a sufficient recombiner estimator when analyzer asymmetry is simultaneously large. A perturbative coupled estimator based on $(H_2, H_3, \Delta\phi_{13})$ jointly can improve recovery, but full two-parameter estimation across the entire severity grid requires local calibration beyond the scope of this initial framework.

Comparison with direct model fitting. A natural alternative to the phase-descent test is to fit a parametric error model — say, $\delta(b) = A_1 \cos(b - \alpha_1) + A_3 \cos(3b - \alpha_3)$ — directly to the observed angular sweep, then read off fault attribution from the recovered amplitudes and phases. This works, but it presupposes the error structure rather than testing it. Three operational differences favor the phase-coherence formulation. *First*, phase descent is a necessary condition that *any* single-origin distortion must satisfy, regardless of which model one would have written down; a fit cannot detect that no single-origin model is adequate, only that a particular chosen model fits poorly. *Second*, the test requires no model selection step: the spectrum of $\delta(b)$ is computed once, and the phase invariant is checked harmonic-by-harmonic without committing to a parametric form. *Third*, the leakage ratio Λ_{53} provides a built-in trust monitor for the linear regime where the perturbative decomposition is meaningful — fitting-based approaches typically need a separate residual-analysis step to provide the equivalent. The two approaches are complementary: phase descent first, parametric fit second once attribution is established.

7 Conclusion

We have established a spectral framework for systematic error classification in passive correlators based on the phase-descent invariant: the algebraic constraint that harmonics generated by a single perturbation through a single nonlinearity satisfy $\phi_n - n\phi_1 = 0 \pmod{\pi}$.

The main contributions are:

1. **The phase-descent invariant and its converse**, which together provide a necessary condition for single-origin errors: phase descent must hold if the distortion is generated by one latent phase variable through one nonlinearity, and its violation rules out that description. While the underlying spectral fact is a consequence of the classical Chebyshev expansion, its use as a fault-lineage diagnostic is, to our knowledge, new.
2. **A two-family error taxonomy** for bright/dark correlators: analyzer asymmetry (phase-locked odd harmonics) versus recombiner asymmetry (phase-incoherent even harmonics plus DC).
3. **A five-step calibration diagnostic** that extracts fault type, fault severity, and its own validity bound from a single angular sweep.
4. **Numerical validation** demonstrating phase-locking to 10^{-4} rad for analyzer errors up to 20% severity, continuous phase drift for recombiner errors, and confirmation that the leakage-ratio trust monitor (Step 5) fires at the perturbative-validity boundary under simultaneous mixed-case perturbations (Figure 3).

The framework provides a spectral method for systematic error classification that should apply naturally to other instruments using normalized-contrast correlation, whenever the observable is well modeled as a function of a single angular variable. The underlying principle — that phase coherence among harmonics is the signature of a single generating mechanism — is not specific to the correlator studied here, though its application to other architectures remains to be validated.

Appendix A Proof of the Phase-Descent Lemma and Corollary

We restate the Lemma from §2.2 and give a self-contained proof, then derive the phase-descent invariant as an immediate corollary.

Lemma (restated). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be analytic on $[-1, 1]$ — that is, f extends to a function holomorphic on an open neighborhood of $[-1, 1]$, with power series $f(u) = \sum_{k=0}^{\infty} a_k u^k$ converging on some open disk containing $[-1, 1]$. Define $g(\theta) = f(\cos(\theta - \theta_0))$. Then g has a uniformly convergent Fourier cosine expansion

$$g(\theta) = \sum_{n=0}^{\infty} c_n \cos(n(\theta - \theta_0))$$

with all $c_n \in \mathbb{R}$.

Proof. Let $\psi = \theta - \theta_0$. By assumption the power series $\sum_k a_k u^k$ converges on a disk strictly containing $[-1, 1]$, hence converges absolutely and uniformly on $[-1, 1]$ by Abel's theorem. Composing with the bounded continuous map $\psi \mapsto \cos \psi \in [-1, 1]$, the series $\sum_k a_k \cos^k \psi$ converges uniformly in ψ , so we may interchange summation with Fourier projection.

Each power $\cos^k \psi$ admits the standard binomial-Chebyshev expansion

$$\cos^k \psi = \frac{1}{2^k} \sum_{j=0}^k \binom{k}{j} e^{i(k-2j)\psi} = \begin{cases} \frac{2}{2^k} \sum_{j=0}^{(k-1)/2} \binom{k}{j} \cos((k-2j)\psi), & k \text{ odd}, \\ \frac{1}{2^k} \binom{k}{k/2} + \frac{2}{2^k} \sum_{j=0}^{k/2-1} \binom{k}{j} \cos((k-2j)\psi), & k \text{ even}, \end{cases}$$

which follows from $\cos \psi = (e^{i\psi} + e^{-i\psi})/2$ and the binomial theorem, then pairing j and $k-j$ terms. Each cosine $\cos(m\psi)$ on the right has $m \leq k$ and $m \equiv k \pmod{2}$, and (for k even) the constant term contributes only to the $n = 0$ harmonic.

Substituting and collecting terms with $m = n$:

$$c_n = \sum_{\substack{k \geq n \\ k \equiv n \pmod{2}}} a_k \cdot \kappa_{k,n}, \quad \kappa_{k,n} = \begin{cases} 2^{1-k} \binom{k}{(k-n)/2} & n \geq 1, \\ 2^{-k} \binom{k}{k/2} & n = 0, \text{ } k \text{ even}, \end{cases}$$

each a finite real combinatorial factor. Uniform convergence justifies the term-by-term collection, so $c_n \in \mathbb{R}$ for every n , and $g(\theta) = \sum_n c_n \cos(n(\theta - \theta_0))$ converges uniformly. $\square$

Corollary (Phase descent, restated). The complex Fourier coefficients H_n of g satisfy

$$\phi_n = \begin{cases} n\theta_0 \pmod{2\pi} & \text{if } c_n > 0, \\ n\theta_0 + \pi \pmod{2\pi} & \text{if } c_n < 0, \end{cases}$$

so $\phi_n - n\phi_1 \equiv 0 \pmod{\pi}$ for all n with $c_n \neq 0$. $\square$

Remark. The proof uses only that $T_n(\cos \psi) = \cos(n\psi)$ — the defining property of Chebyshev polynomials of the first kind. The same argument applies to any function expanded in the Chebyshev basis $\{T_n\}$, regardless of the particular nonlinearity.

Appendix B Network Parameters

The baseline network parameters used in all simulations are listed below.

Table 3: Baseline network parameters used in all simulations.

Self-inductances:		
Source primaries:	Lp	= 1.0 mH
Clarke pickups:	Lalpha,Lbeta	= 0.8 mH
Analyzer coils:	LA,LB	= 0.6 mH
Tank coils:	LBr,LDk	= 1.0 mH
Series resistances:		
Source:	Rp	= 0.5 ohm
Internal coils:		0.2 ohm each
Coupling scales:		
Source→Clarke:	kC	= 0.15
Clarke→Analyzer:	kA	= 0.20
Analyzer→Tank:	kR	= 0.25
Tank terminations:		
Shunt resistance:	R_sh	= 50 ohm
Shunt capacitance:	CBr	= 253.30 nF (tuned to 10 kHz)
Intermediate-port:		
load:	R_kL	= 1000 ohm
Source:		
Frequency:	f	= 10 kHz
Voltage Amplitude:	Vm	= 10 V

Appendix C Notation

Symbol	Definition
$E(a, b)$	Normalized correlator: $(P_{\text{Br}} - P_{\text{Dk}})/(P_{\text{Br}} + P_{\text{Dk}})$
$\delta(b)$	Angular error signal: $E(0, b) - \cos(b)$
H_n	Complex n -th Fourier harmonic of $\delta(b)$
ϕ_n	Phase of H_n : $\arg(H_n)$
ϵ_A	Analyzer asymmetry parameter (fractional α/β imbalance)
ϵ_R	Recombiner asymmetry parameter (fractional branch-local imbalance)
$\Delta\phi_{13}$	Phase-lock quantity: $\text{wrap}(\phi_3 - 3\phi_1)$
Λ_{53}	Leakage ratio: $ H_5 / H_3 $
$\mathbf{L}$	9×9 mutual-inductance matrix
k_C, k_A, k_R	Coupling scales: source→Clarke, Clarke→analyzer, analyzer→tank

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